NOTES

Tutorial → 2

1. Pancake Problem.

→ Most efficient one is Bi directional Sort 3 cake using 5 flips only.

Stack of 'n' pancakes of distinct sizes.

Operation = flip (k)

Goal: Sort the stack (smallest top)

Algorithm:-

place Pancakes one over another starting from Largest (n^{th} position) down to Smallest position (1).

For each pancake 'i' from n down to 2.

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```
Pancake sort (A, n)
    for i = n down to 2;
    max = Find - Max (A, 1, i)
    if max != 1;
        flip (A, max)
        flip (A, i).
Find Max (A, lo, hi)
        max = lo
    for j = Lo + 1 to hi
        if A[j] > A[max]
            max = j
        return Max.

Flip (A, K)
  Left = 1          right = K
        While left < right
```

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swap [A[left]. A[right]]

left ++, right --.

Worst case flips = $2(n-1)$

Time complexity = $\Theta(n^2)$

Comparison	Worst	Time
Simple selection	$2(n-1)$	$\Theta(n^2)$
Optimized	$\leq 2(n-1)$	$O(n^2)$
Optimal [Chitturi]	$[5n/3]$	$\Theta(n^2)$

Another way is heap but we need to find min. flip so Chitturi is optimal.

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2. Direct Path with Step & jump.

Problem Statement.

'N' nodes numbered 1 to n. You can start at 1 to reach 'n'. At each second you choose

- Step (move from i to i+1)
- Jump (1, 2 . . . . . n)

Minimize time to reach 'n'

Assume time = E_i

Base case $E_n = 0$.

Step → E_i step = $E_{i+1} + 1$

Jump → E_i jump = $1 + \frac{1}{n} \sum_{j=1}^{n} E_j = 1 + \bar{E}$

Optimally

$$1 + \bar{E} < 1 + E_{i+1}$$

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There are many ways to solve

1) Naive Greedy
2) Threshold
3) Dynamic Programming

(Fixed Midpoint)

jump if $i < n/2$ else step.

```
Naive (n)
    current = 1
    time k = 0
    while current != n;
    if current < n/2.
        current = jump(n);
    else
        current = current + 1;
        time = time + 1
    return time.
```

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Jump (n)

return random integer Uniformly from $\{1, 2 \ldots n\}$

Expected Time = $\Theta(n)$.